Logistic regression

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1 Introduction

There are many method to classify elements thanks to features given.

We are looking at the regression method, and study the performances of this method

2 Methodology

- X: a set of features $(n \times d)$
- y: a set of labels for the given features $(n \times h)$
- C: the set of all the classes

2.1 Binary case

In this case, $h = 1$.

We consider that we have only two classes $c_1, c_2 \in C$.

We want to define what is the probability of $p(y|X)$.

To do that, we want to define what is the probability for just one instance:

$$p(y_i = c_1|x_i)$$

As we have 2 classes, we can consider that this probability follow a Bernoulli law. So $y_i = 1$ if $y_i = c_1$ and $y_i = 0$ if $y_i \neq c_1$.

The probability to have 1 is given by a function that depends on x_i. So we have $p(y_i = 1|x_i) = g(x_i)$

So it means that we have:

$$p(y_i|x_i) = \begin{cases} p(y_i = 1|x_i) \text{ if } y_i = c_1 \\ 1 - p(y_i = 1|x_i) \text{ if } y_i \neq c_1 \end{cases} \tag{1}$$

We can write the equation 1 like this:

$$p(y_i|x_i) = p(y_i = 1|x_i)^{y_i}(1 - p(y_i = 1|x_i))^{1-y_i} \tag{2}$$

But we want to know what is the function $g(x_i)$.

According to Bayes theorem, we have:

$$
\begin{aligned}
p(y_i = 1|x_i) &= \frac{p(x_i|y_i = 1)p(y_i = 1)}{p(x_i)} \\
&= \frac{p(x_i|y_i = 1)p(y_i = 1)}{p(x_i|y_i = 1)p(y_i = 1) + p(x_i|y_i = 0)p(y_i = 0)} \\
&= \frac{1}{1 + \frac{p(x_i|y_i=0)p(y_i=0)}{p(x_i|y_i=1)p(y_i=1)}} \\
&= \frac{1}{1 + \frac{p(x_i|y_i=0)p(y_i=0)}{p(x_i|y_i=1)p(y_i=1)}} \\
&= \frac{1}{1 + e^{-a}} \\
&= \sigma(a)
\end{aligned}
\tag{3}
$$

Here, in the equation 3, we consider that we can write $\frac{p(x_i|y_i=0)p(y_i=0)}{p(x_i|y_i=1)p(y_i=1)} = e^{-a}$ with a a scalar.

So we have:

$$
p(y_i = 1|x_i) = \sigma(a) \tag{4}
$$

Now, we want to map all our datas x_i in a scalar.